Telecom / Wireless Communication

Questions 1–50 with Short Calculations and Answers

Q1. A telecom operator wins a spectrum auction for 45 MHz. Company policy sets aside 25 kHz for every simplex channel. How many full-duplex voice channels can this spectrum support?

45 MHz = 45,000 kHz. Full-duplex needs 2 × 25 = 50 kHz.
Channels = 45,000 / 50 = 900.
Answer: **900 full-duplex channels**.

Q2. A regional carrier runs its network on 560 total voice channels, shared across a 7-cell frequency reuse cluster. How many channels are allocated to a single cell?

Channels/cell = 560 / 7 = **80**.
Answer: **80 channels/cell**.

Q3. A GSM operator sets carrier spacing at 200 kHz and rolls out 100 carriers, each carrying 8 time slots for voice traffic. Determine the total number of simultaneous voice channels.

Total voice channels = 100 × 8 = **800**.
Answer: **800 simultaneous channels**.

Q4. A base station radiates 80 W while a nearby handset transmits 4 W. Express the power gap as a decibel-based power control ratio.

10 log10(80/4) = 10 log10(20) = **13.01 dB**.
Answer: **13.01 dB**.

Q5. A serving signal is -60 dBm and co-channel interference is -92 dBm. Find C/I in dB.

C/I = -60 – (–92) = **32 dB**.
Answer: **32 dB**.

Q6. An operator holds 42 MHz of spectrum, assigning 300 kHz per channel, and reuses frequencies across a 7-cell pattern. How many channels does a single cell get?

Total channels = 42,000/300 = 140.
Per cell = 140/7 = **20**.
Answer: **20 channels/cell**.

Q7. A transmitter delivers 64 W to a receiver. The receiver is moved twice as far away, with path-loss exponent 4. What received power is expected?

Power scales as 1/d■. Doubling distance gives 64/2■ = 64/16 = **4 W**.
Answer: **4 W**.

Q8. Cell radius is doubled while keeping the same SNR at the new edge under an inverse-square power law. By what factor should transmit power be scaled?

$P \propto d^2$. If distance doubles: scale = 2^2 = **4**.
Answer: **4× transmit power**.

Q9. A network has a cluster size of 4, with 15 voice channels per cell. Find the total channel count across the cluster.

4 × 15 = **60 channels**.
Answer: **60 channels**.

Q10. A vehicle travels at 100 km/h on a 900 MHz carrier. Estimate the Doppler shift.

v = 100/3.6 = 27.78 m/s.
fD = vf/c = (27.78 × 900×10■)/(3×10■) = **83.3 Hz**.
Answer: **83.3 Hz**.

Q11. A TDMA frame lasts 5 ms and is divided into 8 equal time slots. Find one slot duration.

5/8 = **0.625 ms**.
Answer: **0.625 ms**.

Q12. There are 24 MHz of bandwidth with 200 kHz carrier spacing. How many carrier frequencies can be created?

24,000/200 = **120**.
Answer: **120 carriers**.

Q13. A handset transmits at 8 W and experiences 33 dB path loss. Estimate received power in dBm.

8 W = 8000 mW; 10log10(8000) = 39.03 dBm.
Received = 39.03 – 33 = **6.03 dBm**.
Answer: **6.03 dBm**.

Q14. Using 1 m as reference distance, estimate free-space path loss at 2.5 km for a 1900 MHz deployment.

FSPL(dB) = 32.44 + 20log10(fMHz) + 20log10(dkm).
= 32.44 + 20log10(1900) + 20log10(2.5) ≈ **105.98 dB**.
Answer: ≈**106 dB**.

Q15. A cluster has 6 cells, each carrying up to 250 simultaneous calls. Find cluster capacity.

6 × 250 = **1,500 calls**.
Answer: **1,500 simultaneous calls**.

Q16. A switching subsystem processes 4000 new call requests per second, with average call duration 3 minutes. How many calls could be active at once?

3 min = 180 s.
Active calls = 4000 × 180 = **720,000**.
Answer: **720,000 active calls**.

Q17. 150 channels are assigned to a base station, each occupying 25 kHz. Find total bandwidth.

150 × 25 kHz = 3750 kHz = **3.75 MHz**.
Answer: **3.75 MHz**.

Q18. A 40 kHz guard band is inserted between 30 frequency bands. How much bandwidth is wasted?

30 bands create 29 gaps between them.
29 × 40 kHz = 1160 kHz = **1.16 MHz**.
Answer: **1.16 MHz**.

Q19. A vehicle travels at 90 km/h through cells of radius 1.5 km. Using cell diameter as the crossing distance, how many handoffs occur in one hour?

Diameter = 2 × 1.5 = 3 km.
Handoffs/hour ≈ 90/3 = **30**.
Answer: **30 handoffs/hour**.

Q20. A carrier network has 8 regional centers, each overseeing 40 base stations, with each base station serving 250 users. Find total users.

8 × 40 × 250 = **80,000 users**.
Answer: **80,000 users**.

Q21. A region of 400 km² is covered using hexagonal cells of 400 m radius. Approximate how many cells are required?

R = 0.4 km. Hexagonal area ≈ $(3\sqrt{3}/2)R^2$ = 0.4157 km².
Cells ≈ 400/0.4157 = 962.25, so round up to **963 cells** to cover the region.

Q22. An FDMA system uses 30 kHz per channel from a 6 MHz spectrum pool. How many channels are available?

6,000/30 = **200 channels**.
Answer: **200 channels**.

Q23. There are 1200 subscribers, each making 4 calls/hour with average duration 2.5 minutes. Find offered traffic in Erlangs.

Traffic = 1200 × 4 × (2.5/60) = **200 Erlangs**.
Answer: **200 Erlangs**.

Q24. A 2.5 GHz deployment has 40 MHz bandwidth, and each connected user consumes 400 kHz. Find maximum simultaneous users.

40,000/400 = **100 users**.
Answer: **100 simultaneous users**.

Q25. Each of 1200 base stations supports 600 simultaneous users. Find total network capacity.

1200 × 600 = **720,000 users**.
Answer: **720,000 simultaneous users**.

Q26. A paging scheme sweeps 4 cells at a time across a 200-cell network. In the worst case, how many paging rounds are needed?

Rounds = 200/4 = **50 rounds**.
Answer: **50 paging rounds**.

Q27. A network supports 100,000 simultaneous connections. It receives 40,000 call attempts/minute, with average duration 3 minutes. What fraction of offered traffic is blocked?

Offered traffic = 40,000 × 3 = 120,000 Erlangs-equivalent calls.
Capacity = 100,000. Blocked = 20,000.
Blocked fraction = 20,000/120,000 × 100 = **16.67%**.
Answer: ≈**16.67%**.

Q28. A network policy limits 16% of 600,000 subscribers from entering traffic each hour. How many subscribers are affected?

600,000 × 0.16 = **96,000**.
Answer: **96,000 subscribers**.

Q29. A switching center can process 12,000 calls/s and is handling 9,000 calls/s at peak. Find spare capacity.

12,000 – 9,000 = **3,000 calls/s**.
Answer: **3,000 calls/s spare**.

Q30. Frequencies are reused with factor 4 across 48 MHz of spectrum, with 200 kHz per channel. How many channels does each cell get?

Total channels = 48,000/200 = 240.
Per cell = 240/4 = **60 channels**.
Answer: **60 channels/cell**.

Q31. A 900 MHz GSM deployment uses 200 kHz channel spacing across 30 MHz. How many channels are supported?

30,000/200 = **150 channels**.
Answer: **150 channels**.

Q32. A network records 2500 handoff attempts, with 99% successful. How many calls/handoffs succeed?

2500 × 0.99 = **2475 successful handoffs**.
Answer: **2,475**.

Q33. A mobile handset radiates 12 W and experiences 45 dB path loss. What power does the base station receive?

12 W = 12,000 mW.
10log10(12,000) ≈ 40.79 dBm.
Received = 40.79 – 45 = –**4.21 dBm**.
Answer: –**4.21 dBm**.

Q34. A 7-cell cluster has up to 900 simultaneous users per cell. Find cluster capacity.

7 × 900 = **6,300 users**.
Answer: **6,300 simultaneous users**.

Q35. A TDMA frame spans 4.8 ms and is split into 8 equal slots. Find slot duration.

4.8/8 = **0.6 ms**.
Answer: **0.6 ms**.

Q36. A mobile moves at 150 km/h connected to a 1.8 GHz carrier. Estimate Doppler shift.

v = 150/3.6 = 41.67 m/s.
fD = vf/c = (41.67 × 1.8×10■)/(3×10■) = **250 Hz**.
Answer: **250 Hz**.

Q37. A core switch supports 250,000 simultaneous calls. Call arrival is 70,000/min and average duration is 4 min. What share of offered load is blocked?

Offered load = 70,000 × 4 = 280,000.
Blocked = 280,000 – 250,000 = 30,000.
Blocked share = 30,000/280,000 ×100 ≈ **10.71%**.
Answer: ≈**10.71%**.

Q38. A hexagonal layout contains 9 cells, each with 600 users. Find cluster capacity.

9 × 600 = **5,400 users**.
Answer: **5,400 users**.

Q39. A switching center processes 60,000 calls/s, with 0.4% dropped. How many calls are dropped each second?

60,000 × 0.004 = **240 calls/s**.
Answer: **240 calls/s**.

Q40. A base station transmits at 43 dBm and the received signal is −87 dBm. Find path loss.

Path loss = 43 – (–87) = **130 dB**.
Answer: **130 dB**.

Q41. One base station covers 2.5 km², and 150 stations are deployed. Find total geographic coverage.

150 × 2.5 = **375 km²**.
Answer: **375 km²**.

Q42. A frequency reuse factor of 6 is used over 48 MHz, with each channel taking 200 kHz. How many channels does each cell get?

Total channels = 48,000/200 = 240.
Per cell = 240/6 = **40 channels**.
Answer: **40 channels/cell**.

Q43. A paging system processes 6000 users per cycle, and the subscriber base is 150,000. How many paging cycles are needed?

150,000/6,000 = **25 cycles**.
Answer: **25 paging cycles**.

Q44. A phone call lasts 8 minutes, and the network hands the call off roughly every 3 minutes. How many handoffs occur?

8/3 ≈ **2.67**. This corresponds to roughly **3 handoffs** over the call.
Answer: ≈**3 handoffs**.

Q45. A receiver gets a wanted signal of −85 dBm against a noise floor of −115 dBm. Find SNR.

SNR = –85 – (–115) = **30 dB**.
Answer: **30 dB**.

Q46. A 4G cell serves 250 users simultaneously, each needing 150 kbps. Find minimum bandwidth.

250 × 150 kbps = 37,500 kbps = **37.5 Mbps**.
Answer: **37.5 Mbps**.

Q47. A base station handles 1500 subscribers, each averaging 4 calls/hour with a 2.5-minute call length. Find traffic in Erlangs.

Traffic = 1500 × 4 × (2.5/60) = **250 Erlangs**.
Answer: **250 Erlangs**.

Q48. A network reserves 250 control channels, each 12 kHz wide. Find total control-channel bandwidth.

250 × 12 kHz = 3000 kHz = **3 MHz**.
Answer: **3 MHz**.

Q49. A network carries 1500 simultaneous connections, each streaming at 600 kbps on average. Find total throughput.

1500 × 600 kbps = 900,000 kbps = **900 Mbps**.
Answer: **900 Mbps**.

Q50. Each tower covers a 4 km radius (circle), and a 350 km² service region must be blanketed. How many towers are required?

Area/tower = $\pi r^2 = \pi(4^2) = 50.27$ km².
Towers = 350/50.27 = 6.96, so round up.
Answer: **7 towers**.